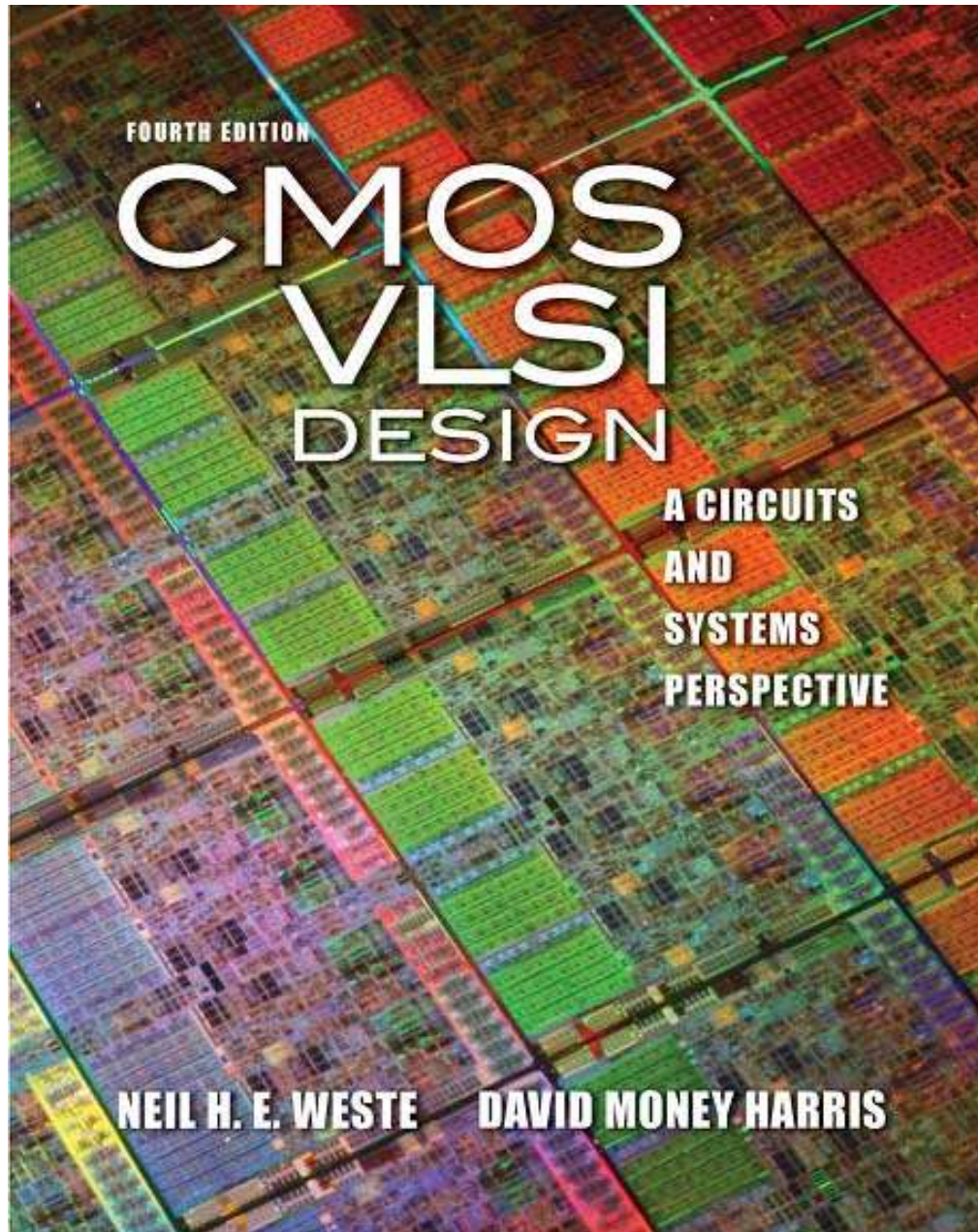




Lecture 4: Nonideal Transistor Theory

Outline

- ❑ Nonideal Transistor Behavior
 - High Field Effects
 - Mobility Degradation
 - Velocity Saturation
 - Channel Length Modulation
 - Threshold Voltage Effects
 - Body Effect
 - Drain-Induced Barrier Lowering
 - Short Channel Effect
 - Leakage
 - Subthreshold Leakage
 - Gate Leakage
 - Junction Leakage
- ❑ Process and Environmental Variations

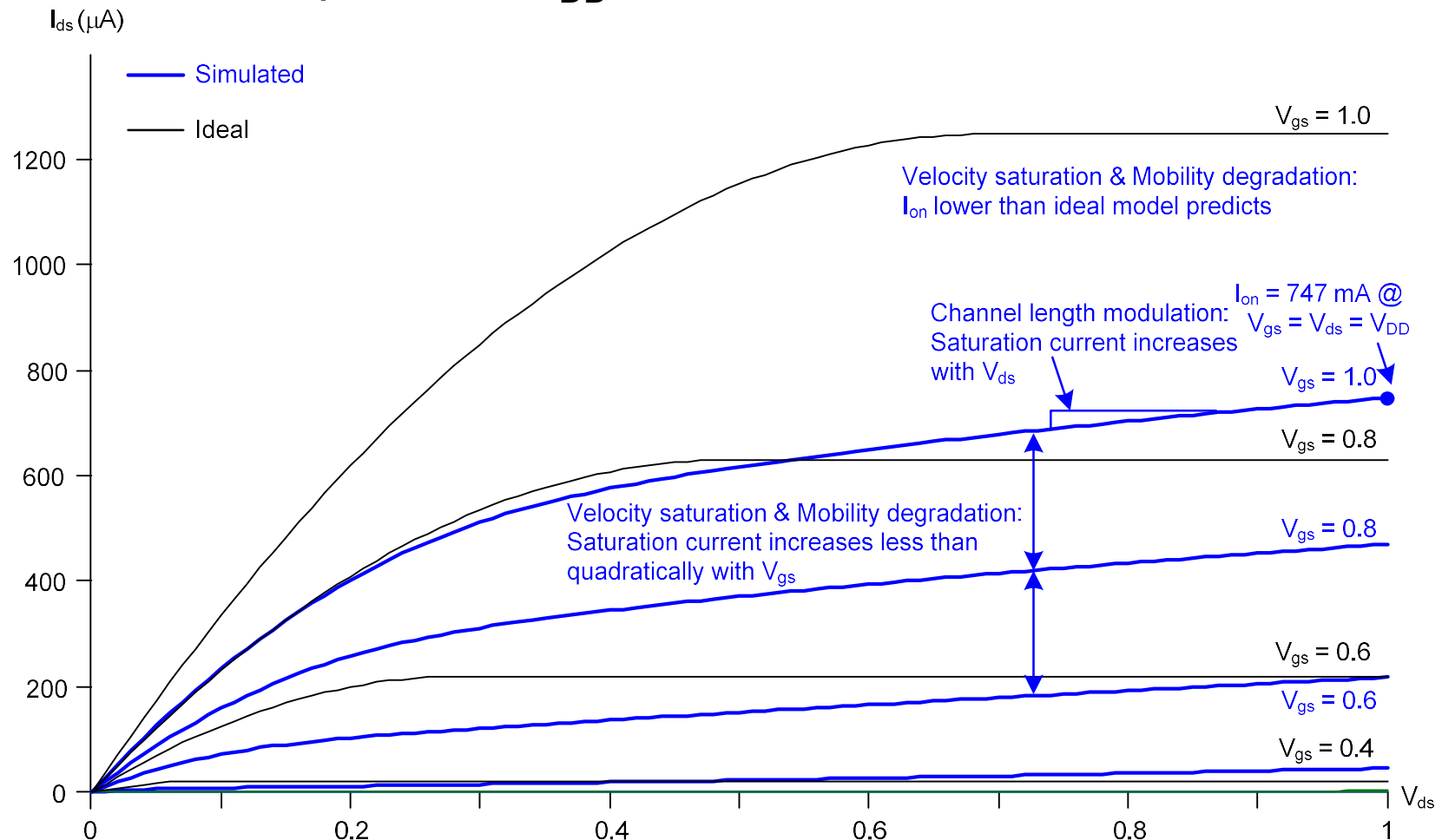
Ideal Transistor I-V

- Shockley long-channel transistor models

$$I_{ds} = \begin{cases} 0 & V_{gs} < V_t & \text{cutoff} \\ \beta \left(V_{gs} - V_t - \frac{V_{ds}}{2} \right) V_{ds} & V_{ds} < V_{dsat} & \text{linear} \\ \frac{\beta}{2} (V_{gs} - V_t)^2 & V_{ds} > V_{dsat} & \text{saturation} \end{cases}$$

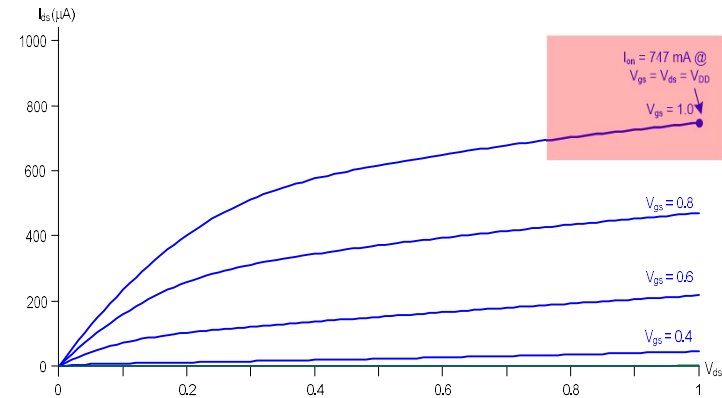
Ideal vs. Simulated nMOS I-V Plot

□ 65 nm IBM process, $V_{DD} = 1.0$ V

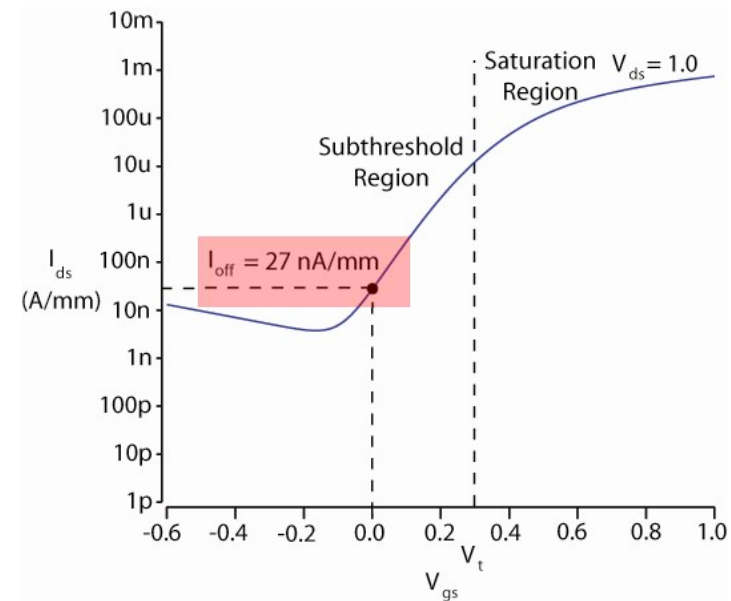


ON and OFF Current

□ $I_{on} = I_{ds} @ V_{gs} = V_{ds} = V_{DD}$
– Saturation



□ $I_{off} = I_{ds} @ V_{gs} = 0, V_{ds} = V_{DD}$
– Cutoff



Electric Fields Effects

- Vertical electric field: $E_{\text{vert}} =$ _____
 - Attracts carriers into channel
 - Long channel: $Q_{\text{channel}} \propto E_{\text{vert}}$
- Lateral electric field: $E_{\text{lat}} =$ _____
 - Accelerates carriers from drain to source
 - Long channel: $v = \mu E_{\text{lat}}$

Coffee Cart Analogy

- ❑ Tired student runs from VLSI lab to coffee cart
- ❑ Freshmen are pouring out of the physics lecture hall
- ❑ V_{ds} is how long you have been up
 - Your velocity = fatigue $\times$ mobility
- ❑ V_{gs} is a wind blowing you against the glass (SiO_2) wall
- ❑ At high V_{gs} , you are buffeted against the wall
 - *Mobility degradation*
- ❑ At high V_{ds} , you scatter off freshmen, fall down, get up
 - *Velocity saturation*
 - Don't confuse this with the saturation region

Mobility Degradation

- High E_{vert} effectively reduces mobility
 - Collisions with oxide interface

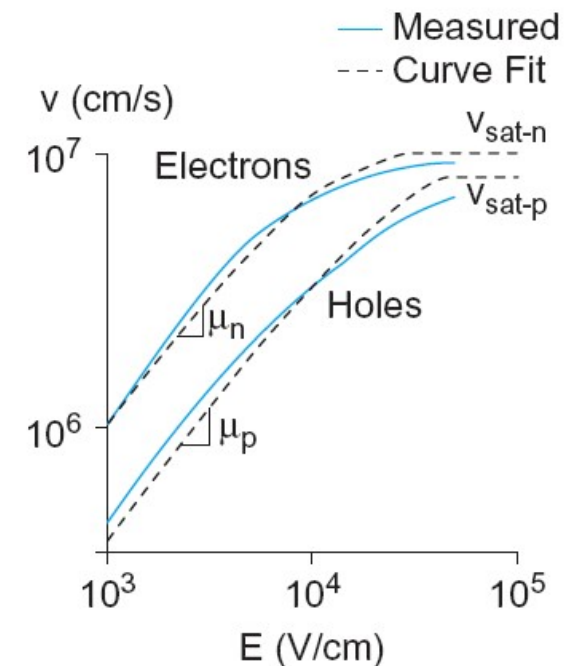
$$\mu_{\text{eff}-n} = \frac{540 \frac{\text{cm}^2}{\text{V} \cdot \text{s}}}{1 + \left(\frac{V_{gs} + V_t}{0.54 \frac{\text{V}}{\text{nm}} t_{\text{ox}}} \right)^{1.85}}$$

$$\mu_{\text{eff}-p} = \frac{185 \frac{\text{cm}^2}{\text{V} \cdot \text{s}}}{1 + \frac{\left| V_{gs} + 1.5V_t \right|}{0.338 \frac{\text{V}}{\text{nm}} t_{\text{ox}}}}$$

Velocity Saturation

- At high E_{lat} , carrier velocity rolls off
 - Carriers scatter off atoms in silicon lattice
 - Velocity reaches v_{sat}
 - Electrons: 10^7 cm/s
 - Holes: 8×10^6 cm/s
 - Better model

$$v = \begin{cases} \frac{\mu_{eff} E}{1 + \frac{E}{E_c}} & E < E_c \\ v_{sat} & E \geq E_c \end{cases} \quad E_c = \frac{2v_{sat}}{\mu_{eff}}$$



Vel Sat I-V Effects

- ❑ Ideal transistor ON current increases with V_{DD}^2

$$I_{ds} = \mu C_{ox} \frac{W}{L} \frac{(V_{gs} - V_t)^2}{2} = \frac{\beta}{2} (V_{gs} - V_t)^2$$

- ❑ Velocity-saturated ON current increases with V_{DD}

$$I_{ds} = C_{ox} W (V_{gs} - V_t) v_{max}$$

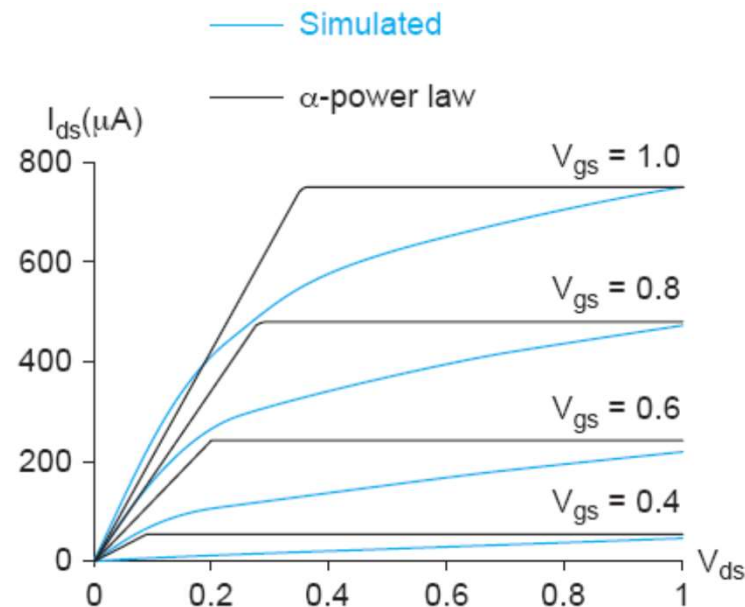
- ❑ Real transistors are partially velocity saturated
 - Approximate with α -power law model
 - $I_{ds} \propto V_{DD}^\alpha$
 - $1 < \alpha < 2$ determined empirically (≈ 1.3 for 65 nm)

α -Power Model

$$I_{ds} = \begin{cases} 0 & V_{gs} < V_t & \text{cutoff} \\ I_{dsat} \frac{V_{ds}}{V_{dsat}} & V_{ds} < V_{dsat} & \text{linear} \\ I_{dsat} & V_{ds} > V_{dsat} & \text{saturation} \end{cases}$$

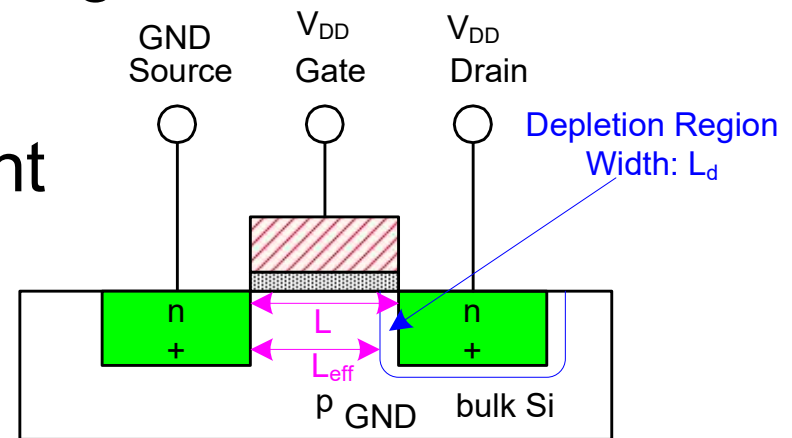
$$I_{dsat} = P_c \frac{\beta}{2} (V_{gs} - V_t)^\alpha$$

$$V_{dsat} = P_v (V_{gs} - V_t)^{\alpha/2}$$



Channel Length Modulation

- ❑ Reverse-biased p-n junctions form a *depletion region*
 - Region between n and p with no carriers
 - Width of depletion L_d region grows with reverse bias
 - $L_{\text{eff}} = L - L_d$
- ❑ Shorter L_{eff} gives _____ current
 - I_{ds} _____ with V_{ds}
 - Even in saturation



Chan Length Mod I-V

$$I_{ds} = \frac{\beta}{2} (V_{gs} - V_t)^2 (1 + \lambda V_{ds})$$

- λ = *channel length modulation coefficient*
 - not feature size
 - Empirically fit to I-V characteristics

Threshold Voltage Effects

- ❑ V_t is V_{gs} for which the channel starts to invert
- ❑ Ideal models assumed V_t is constant
- ❑ Really depends (weakly) on almost everything else:
 - Body voltage: *Body Effect*
 - Drain voltage: *Drain-Induced Barrier Lowering*
 - Channel length: *Short Channel Effect*

Body Effect

- ❑ Body is a fourth transistor terminal
- ❑ V_{sb} affects the charge required to invert the channel
 - Increasing V_s or decreasing V_b increases V_t

$$V_t = V_{t0} + \gamma \left(\sqrt{\phi_s + V_{sb}} - \sqrt{\phi_s} \right)$$

- ❑ $\phi_s = \text{surface potential at threshold}$

$$\phi_s = 2v_T \ln \frac{N_A}{n_i}$$

- Depends on doping level N_A
- And intrinsic carrier concentration n_i

- ❑ $\gamma = \text{body effect coefficient}$

$$\gamma = \frac{t_{ox}}{\epsilon_{ox}} \sqrt{2q\epsilon_{si}N_A} = \frac{\sqrt{2q\epsilon_{si}N_A}}{C_{ox}}$$

Body Effect Cont.

- For small source-to-body voltage, treat as linear

$$V_t = V_{t0} + k_\gamma V_{sb}$$

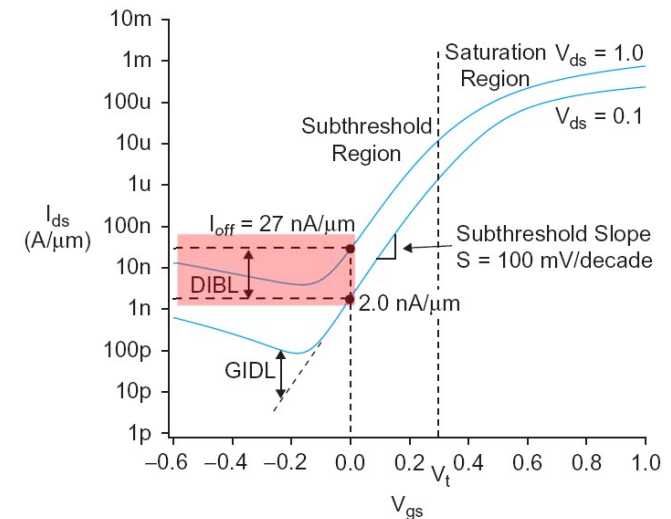
$$k_\gamma = \frac{\gamma}{2\sqrt{\phi_s}} = \frac{\sqrt{\frac{q\epsilon_{si}N_A}{v_T \ln \frac{N_A}{n_i}}}}{2C_{ox}}$$

DIBL

- ❑ Electric field from drain affects channel
- ❑ More pronounced in small transistors where the drain is closer to the channel
- ❑ Drain-Induced Barrier Lowering
 - Drain voltage also affect V_t

$$V'_t = V_t - \eta V_{ds}$$

- ❑ High drain voltage causes current to _____



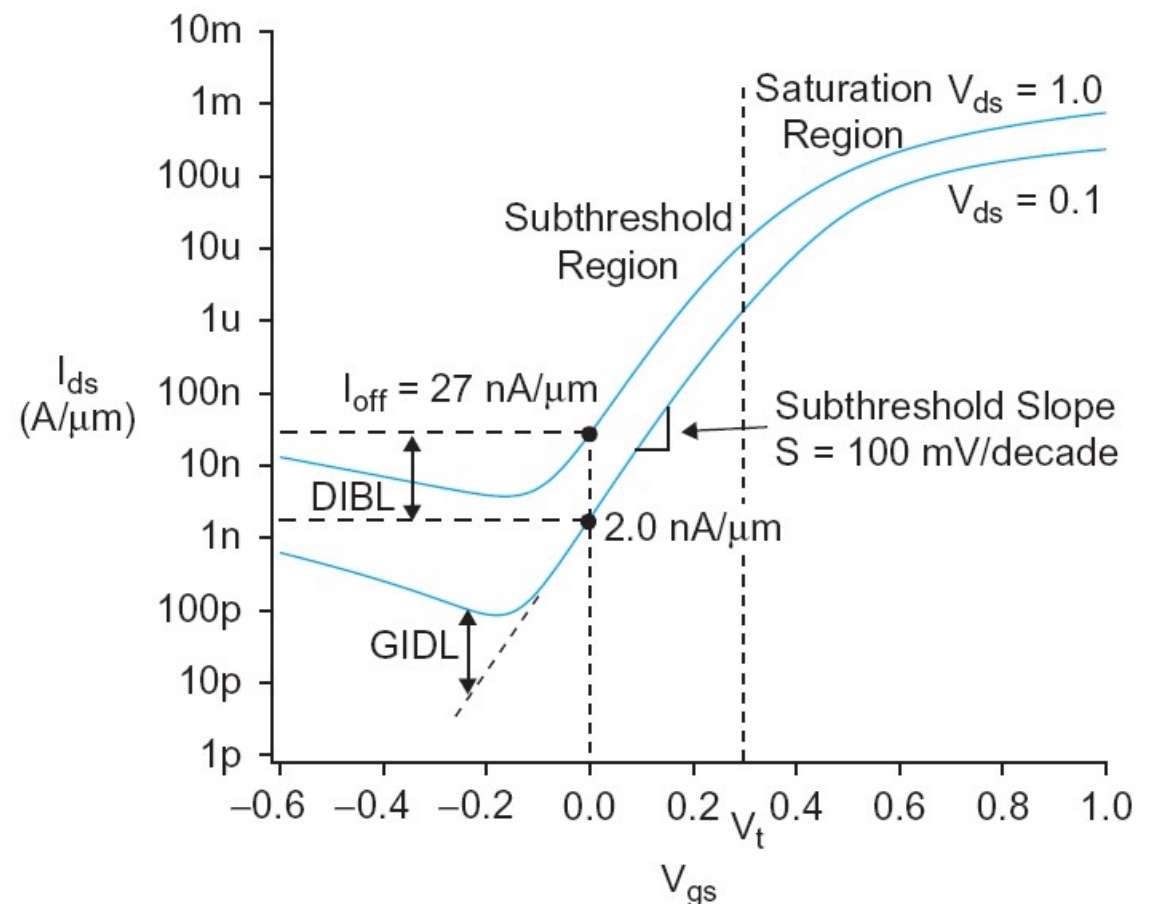
Short Channel Effect

- ❑ In small transistors, source/drain depletion regions extend into the channel
 - Impacts the amount of charge required to invert the channel
 - And thus makes V_t a function of channel length
- ❑ Short channel effect: V_t increases with L
 - Some processes exhibit a reverse short channel effect in which V_t decreases with L

Leakage

- ❑ What about current in cutoff?
- ❑ Simulated results
- ❑ What differs?

—



Leakage Sources

- ❑ Subthreshold conduction
 - Transistors can't abruptly turn ON or OFF
 - Dominant source in contemporary transistors
- ❑ Gate leakage
 - Tunneling through ultrathin gate dielectric
- ❑ Junction leakage
 - Reverse-biased PN junction diode current

Subthreshold Leakage

- Subthreshold leakage exponential with V_{gs}

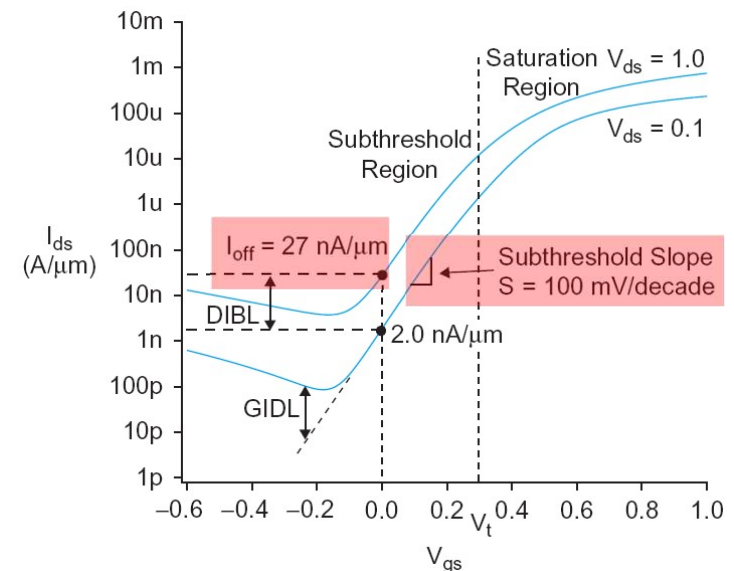
$$I_{ds} = I_{ds0} e^{\frac{V_{gs} - V_{t0} + \eta V_{ds} - k_{\gamma} V_{sb}}{nv_T}} \left(1 - e^{\frac{-V_{ds}}{v_T}} \right)$$

- n is process dependent
 - typically 1.3-1.7
- Rewrite relative to I_{off} on log scale

$$I_{ds} = I_{off} 10^{\frac{V_{gs} + \eta(V_{ds} - V_{dd}) - k_{\gamma} V_{sb}}{S}} \left(1 - e^{\frac{-V_{ds}}{v_t}} \right)$$

$$S = \left[\frac{d(\log_{10} I_{ds})}{dV_{gs}} \right]^{-1} = nv_T \ln 10$$

- $S \approx 100 \text{ mV/decade}$ @ room temperature



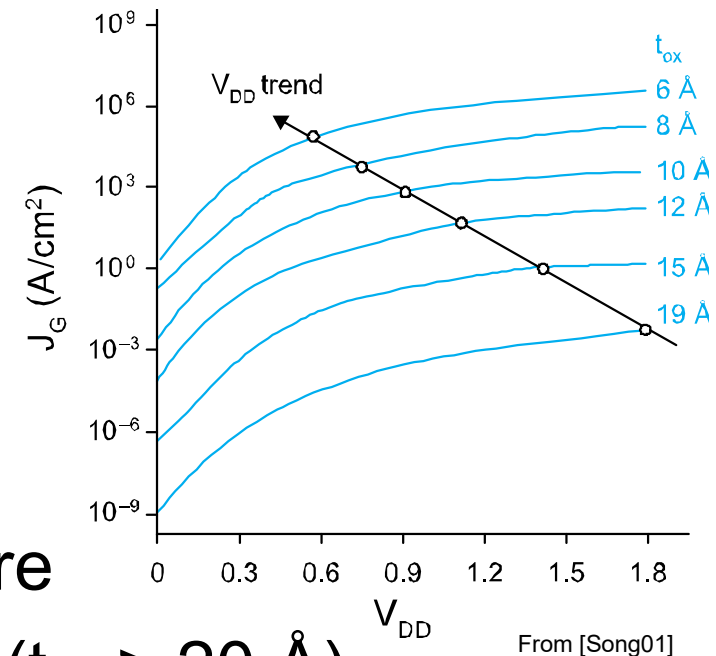
Gate Leakage

- ❑ Carriers tunnel thorough very thin gate oxides
- ❑ Exponentially sensitive to t_{ox} and V_{DD}

$$I_{gate} = WA \left(\frac{V_{DD}}{t_{ox}} \right)^2 e^{-B \frac{t_{ox}}{V_{DD}}}$$

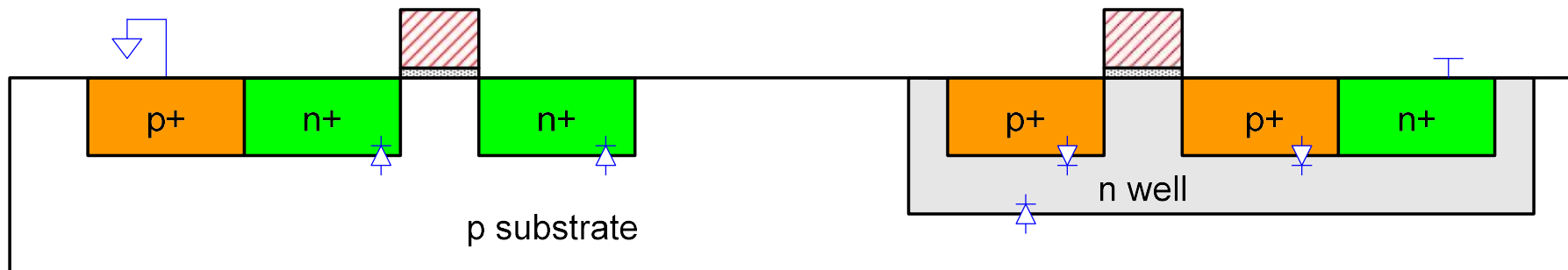
- A and B are tech constants
- Greater for electrons
 - So nMOS gates leak more

- ❑ Negligible for older processes ($t_{ox} > 20 \text{ \AA}$)
- ❑ Critically important at 65 nm and below ($t_{ox} \approx 10.5 \text{ \AA}$)



Junction Leakage

- ❑ Reverse-biased p-n junctions have some leakage
 - Ordinary diode leakage
 - Band-to-band tunneling (BTBT)
 - Gate-induced drain leakage (GIDL)



Diode Leakage

- ❑ Reverse-biased p-n junctions have some leakage

$$I_D = I_S \left(e^{\frac{V_D}{V_T}} - 1 \right)$$

- ❑ At any significant negative diode voltage, $I_D = -I_S$
- ❑ I_S depends on doping levels
 - And area and perimeter of diffusion regions
 - Typically $< 1 \text{ fA}/\mu\text{m}^2$ (negligible)

Band-to-Band Tunneling

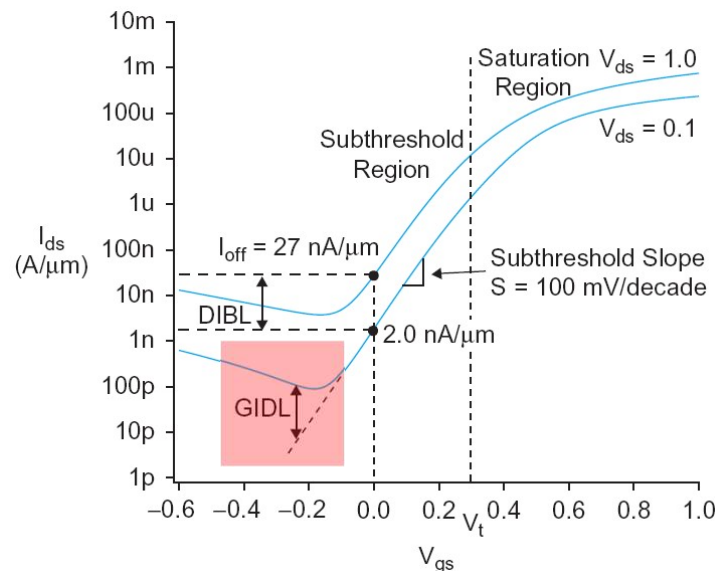
- ❑ Tunneling across heavily doped p-n junctions
 - Especially sidewall between drain & channel when *halo doping* is used to increase V_t
- ❑ Increases junction leakage to significant levels

$$I_{BTBT} = WX_j A \frac{E_j}{E_g^{0.5}} V_{dd} e^{-B \frac{E_g^{1.5}}{E_j}} \quad E_j = \sqrt{\frac{2qN_{halo}N_{sd}}{\epsilon(N_{halo} + N_{sd})}} \left(V_{DD} + v_T \ln \frac{N_{halo}N_{sd}}{n_i^2} \right)$$

- X_j : sidewall junction depth
- E_g : bandgap voltage
- A, B : tech constants

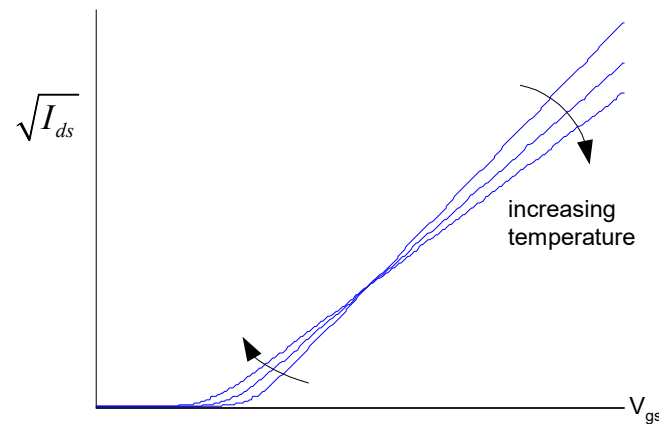
Gate-Induced Drain Leakage

- ❑ Occurs at overlap between gate and drain
 - Most pronounced when drain is at V_{DD} , gate is at a negative voltage
 - Thwarts efforts to reduce subthreshold leakage using a negative gate voltage



Temperature Sensitivity

- ☐ Increasing temperature
 - Reduces mobility
 - Reduces V_t
- ☐ I_{ON} _____ with temperature
- ☐ I_{OFF} _____ with temperature



So What?

- ❑ So what if transistors are not ideal?
 - They still behave like switches.
- ❑ But these effects matter for...
 - Supply voltage choice
 - Logical effort
 - Quiescent power consumption
 - Pass transistors
 - Temperature of operation

Parameter Variation

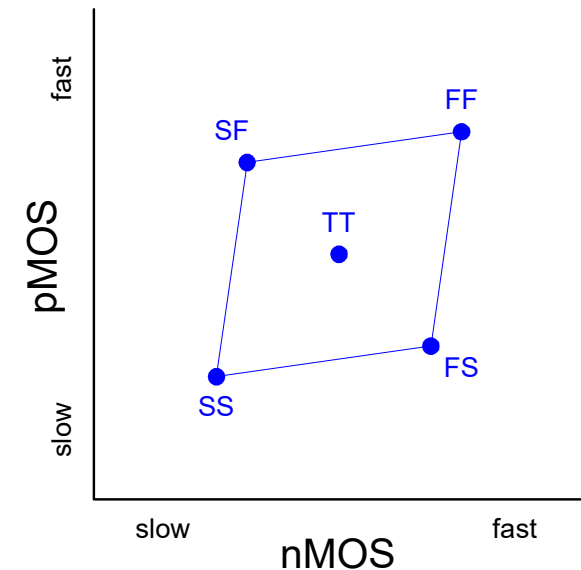
- ❑ Transistors have uncertainty in parameters
 - Process: L_{eff} , V_t , t_{ox} of nMOS and pMOS
 - Vary around typical (T) values

- ❑ Fast (F)

- L_{eff} : _____
- V_t : _____
- t_{ox} : _____

- ❑ Slow (S): opposite

- ❑ Not all parameters are independent for nMOS and pMOS



Environmental Variation

- ☐ V_{DD} and T also vary in time and space
- ☐ Fast:
 - V_{DD} : _____
 - T : _____

Corner	Voltage	Temperature
F		
T	1.8	70 C
S		

Process Corners

- ❑ Process corners describe worst case variations
 - If a design works in all corners, it will probably work for any variation.
- ❑ Describe corner with four letters (T, F, S)
 - nMOS speed
 - pMOS speed
 - Voltage
 - Temperature

Important Corners

- ❑ Some critical simulation corners include

Purpose	nMOS	pMOS	V_{DD}	Temp
Cycle time				
Power				
Subthreshold leakage				